THE UNIVERSITY OF MELBOURNE

FINAL REPORT

The Art of Scientific Computing: Complex Systems

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Chapter 1

Introduction

This project is based on the "Complex Systems" project outline [4].

1.1 Self-Organised Criticality and Scale Invariance

1. The concept of *self-organised criticality* (SOC) was introduced in 1987 by Bak, Tang and Wiesenfeld [1, 2].

2. If we add a grain of sand to the pile, the sand pile can either remain stable or produce an avalanche. The microscopic phenomena is the same in both cases. The size of an avalanche depends on the configuration of the pile, not just the initial disturbance by the dropped grain of sand.

3. The *Gutenberg-Richter law* [3] describes how, for a given region, the number of earthquakes N of a magnitude M are related by

$$N = 10^{a-bM}$$

for some constants a and b. Since magnitude is a logarithmic measure of earthquake size (i.e., seismic energy E is related to magnitude by a relationship of the form $E \propto c^{dM}$ for some constants c and d), it follows that the number of earthquakes is a power law in the energy of the earthquake.

Question 1. Why is it that a system exhibiting scale invariance will have phenomena which are described by a power law?

1.2 Aim

Chapter 2

Bak-Tang-Wiesenfeld Model

Question 2. In what ways does the BTW model satisfy the properties required for a system to exhibit SOC?

Question 3. Why is it important for the table to have finite boundaries? What if we had an infinite lattice, but dropped the sand on only a (finite) subset of the lattice?

Question 4. We can define the average height of each pile to be

$$p_t \coloneqq \frac{1}{NM} \sum_{i,j} p_t(i,j)$$

Without simulating anything, what might you expect this to be once the system has reached a steady state (if it ever does so)? Compare your initial expectation to what your simulation does.

Question 5. Explain the difference between: a steady state, a statistically stationary state, and an equilibrium state. Do you expect the btw model reach an equilibrium state or statistically stationary state? In terms of system observables, how might we characterize this state?

Question 6. Is there a difference between sampling a system over time and sampling an ensemble of systems? If so, under what circumstances? Hint: Look up the property of ergodicity and argue whether or not you believe the btw model is ergodic.

Question 7. Record the frequency with which avalanches have a specific number of topples, area, loss and length. What kind of distribution do you get for each? If possible, can you find a fit for the distribution? (E.g. for a Gaussian, find the mean and width of the distribution; for a power law, find the exponent.)

Question 8. Do you expect any correlation between the four properties we are recording for the avalanche? Justify why/why not, then verify this with your simulation.

2.1 Abelian Property

In this section, we will show that the topples of the BTW model are *abelian*. This means that in an unstable state, the order in which the unstable sites are toppled does change the final stable configuration.

Let $L \subseteq \mathbb{Z}^d$ be the *sand pile lattice*, where $d \geq 1$ is the dimension of the lattice. Corresponding to a configuration of the sand pile p, we define a function $p : L \to \mathbb{Z}_{\geq 0}$, where $p(x)$ is the number of grains of sand on the lattice point x in the configuration p. Define k to be the *stability threshold* (i.e., the pile topples when a lattice point is of height at least k). Then p is a *stable* configuration if $p(x) < k$ for all $x \in L$, and it is *unstable* if $p(x) \geq k$ for at least one $x \in L$.

We define the *toppling matrix* Δ, which has rows and columns indexed by L. For $x, y \in L$, the entry $\Delta(x, y)$ describes the change in height of site y when there is a toppling at site x. The matrix Δ must satisfy the following conditions:

$$\forall x, y \in L \text{ where } x \neq y, \quad \Delta(x, y) = \Delta(y, x) \geq 0 \tag{2.1}$$

$$\forall x \in L, \quad \Delta(x, x) < 0 \tag{2.2}$$

$$\forall x \in L, \quad \sum_{y \in L} \Delta(x, y) \leq 0 \tag{2.3}$$

$$\sum_{x,y \in L} \Delta(x, y) < 0 \tag{2.4}$$

Question 9. For Equation 2.1 to 2.4, explain the significance and justify its necessity.

Equation 2.1 demands that when a site topples, no other site loses any grains. This is consistent with how we defined toppling. Equation 2.1 also insists that there is some symmetry in how toppling works, in the sense that the number of grains that go to a site y when x is toppled should be the same as the number of grains that go to x when y is toppled.

Equation 2.2 means that a site should lose grains when it is toppled. If Equation 2.2 does not hold for a site x, then when the height at x reaches k (which it eventually will do with probability 1), the pile will never be able to reach a stable configuration.

Equation 2.3 can be written as $-\Delta(x, x) \geq \sum_{y \in L \setminus \{x\}} \Delta(x, y)$ for all $x \in L$, meaning that when a site x is toppled, the number of grains gained by other sites is no more than the number of grains which are lost by x (i.e., we are not creating new grains when toppling).

Equation 2.4 demands that there is at least one site where toppling will result in at least one grain leaving the pile. If Equation 2.4 does not hold, the pile will end up enough grains that it can never reach a stable configuration.

In the BTW model, the toppling matrix is defined as follows:

$$\Delta_{\text{BTW}}(x, y) = \begin{cases} -2d & x = y \\ 1 & x \text{ and } y \text{ are nearest neighbours} \\ 0 & \text{else} \end{cases}$$

Question 10. Explain and justify the definition of the toppling matrix in the btw model and verify that it satisfies Equation 2.1 to 2.4. What is the stability threshold k?

The BTW toppling matrix Δ_{BTW} removes $2d$ grains from a toppled site x and adds 1 grain to each nearest neighbour of x. It is also a symmetric matrix, so it clearly satisfies Equation 2.1 and 2.2. Each site in the interior of the pile has $2d$ nearest neighbours, and each site on the boundary of the pile has less than $2d$ nearest neighbours. Therefore for each $x \in L$,

$$\begin{aligned}\sum_{y\in L} \Delta_{\text{BTW}}(x,y) &= \Delta_{\text{BTW}}(x,y) + \sum_{y\in L\setminus\{x\}} \Delta_{\text{BTW}}(x,y) \\ &= -2d + \#\{\text{nearest neighbours of x}\} \\ &\leq 0\end{aligned}$$

Hence Δ_{BTW} also satisfies Equation 2.3. Since at least one site is on the boundary of the pile, we have

$$\sum_{x,y\in L} \Delta_{\text{BTW}}(x,y) = \sum_{x\notin\partial L} \sum_{y\in L\setminus\{x\}} \Delta_{\text{BTW}}(x,y) + \sum_{x\in\partial L} \sum_{y\in L\setminus\{x\}} \Delta_{\text{BTW}}(x,y) < 0$$

so Δ_{BTW} also satisfies Equation 2.4. The stability threshold is $k = 2d$.

For each $x \in L$, we then define the toppling operator T_x, which maps a configuration p to a new configuration $T_x p$, by

$$(T_x p)(y) = \begin{cases} p(y) + \Delta_{\text{BTW}}(x,y) & p(x) \geq k \\ p(y) & \text{else} \end{cases}$$

Question 11. Show that T_x and $T_{x'}$ commute for an unstable configuration p.

Proof. Let $x, x' \in L$ be distinct unstable sites and let $y \in L$ be any site. Then $p(x) \geq k$ and $p(x') \geq k$. It follows that for any $y \in L$, we have

$$\begin{aligned}(T_{x'}p)(y) &= p(y) + \Delta_{\text{BTW}}(x',y) \\ (T_{x'}p)(x) &= p(x) + \Delta_{\text{BTW}}(x',x) \\ (T_x p)(y) &= p(y) + \Delta_{\text{BTW}}(x,y) \\ (T_x p)(x') &= p(x') + \Delta_{\text{BTW}}(x,x')\end{aligned}$$

Since $x \neq x'$, and $p(x)$ and $p(x')$ are both at least k, it follows that $p(x) + \Delta_{\text{BTW}}(x',x) \geq k$ and $p(x') + \Delta_{\text{BTW}}(x,x') \geq k$. Therefore,

$$\begin{aligned}(T_x T_{x'} p)(y) &= \begin{cases} (T_{x'}p)(y) + \Delta_{\text{BTW}}(x,y) & (T_{x'}p)(x) \geq k \\ (T_{x'}p)(y) & \text{else} \end{cases} \\ &= (T_{x'}p)(y) + \Delta_{\text{BTW}}(x,y) \\ &= p(y) + \Delta_{\text{BTW}}(x',y) + \Delta_{\text{BTW}}(x,y) \\ &= (T_{x'}T_x p)(y)\end{aligned}$$

□

Question 12. Show that T is well defined. In other words, show that given some unstable configuration $p, p' = \mathbb{T}p$ is unique (you can't end with two different stable configurations). This is equivalent to the sequence of T_{x_i} required is unique up to re-ordering. Hint: Read the original papers for hints as to how this can be done.

Chapter 3

Implementation of the model

3.1 Testing

Start by testing the avalanche:

```
Test Sandpile 1:
[[3 3 3]
 [3 4 3]
 [3 3 3]]
After avalanche:
[[1 3 1]
 [3 0 3]
 [1 3 1]]
Topples:  10  Area:  9  Total Loss:  12  Length:  2.8284271247461903

Test Sandpile 2:
[[4 3 3 3 3]
 [3 0 3 0 3]
 [3 3 0 0 3]
 [0 0 0 0 3]
 [0 0 0 0 0]]
After avalanche:
[[2 2 2 1 1]
 [2 0 1 3 1]
 [1 1 2 1 1]
 [1 1 0 1 0]
 [0 0 0 0 1]]
Topples:  13  Area:  13  Total Loss:  12  Length:  5.0

The resulting sandpile and avalanche statistics are correct (they
match the calculations which one can do by hand).
```

Run some test simulations:

Listing 3.1: Test simulation

```
start = time.time()
test_simulation(5, 5, 1000, np.random.default_rng(seed=103))
elapsed = time.time() - start
print("Elapsed time: ", elapsed, " seconds for a 5x5 grid
    with 1000 time steps")

start = time.time()
test_simulation(5, 5, 20000, np.random.default_rng(seed=103))
elapsed = time.time() - start
print("Elapsed time: ", elapsed, " seconds for a 5x5 grid
    with 20000 time steps")

start = time.time()
test_simulation(50, 50, 20000, np.random.default_rng(seed
    =103))
elapsed = time.time() - start
print("Elapsed time: ", elapsed, " seconds for a 50x50 grid
    with 20000 time steps")
```

```
Elapsed time:  0.055567  seconds for a 5x5 grid with 1000 time steps
Elapsed time:  1.081542  seconds for a 5x5 grid with 20000 time steps
Elapsed time:  18.475989  seconds for a 50x50 grid with 20000 time steps
```

The model works, but we see that it scales poorly with the size of the grid. We repeatedly call

```
np.argwhere(pile >= unstable_height)
```

We can improve this by only checking the sites which have been changed by a topple. We also need to optimise the way we find the avalanche area/length, as the current method is inefficient.

3.2 Base Model

3.3 Extension

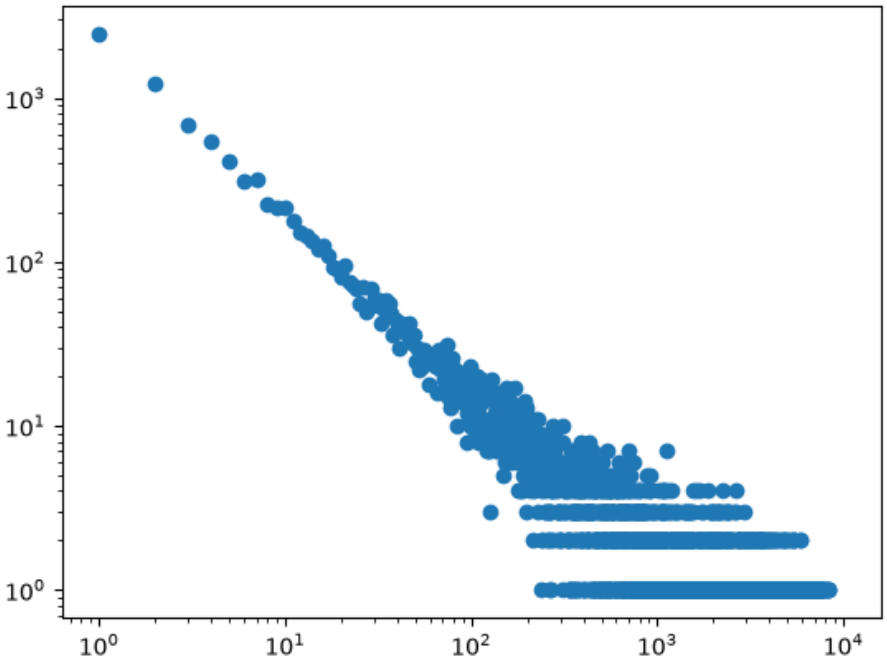


Figure 3.1: area vs frequency

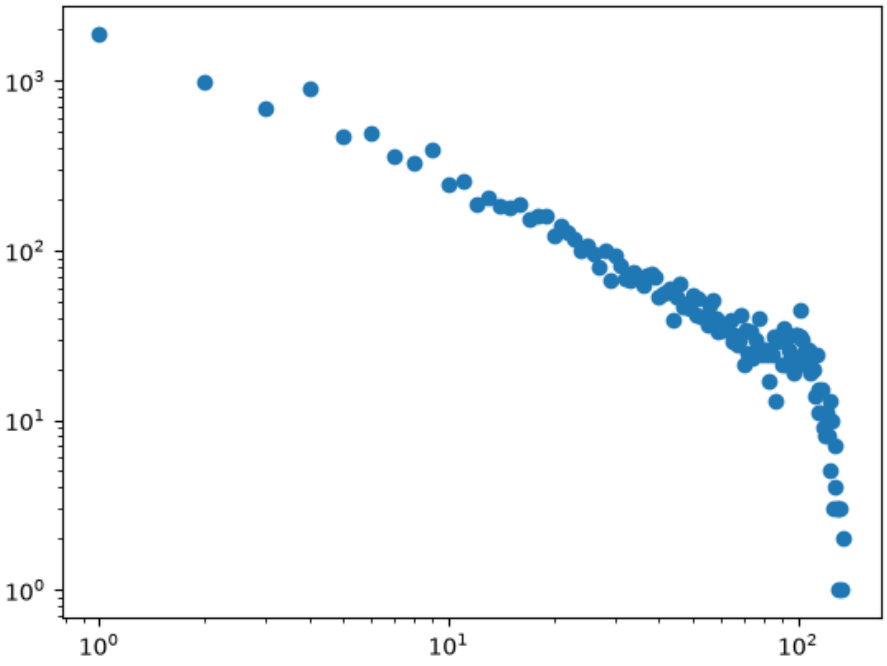


Figure 3.2: length vs frequency

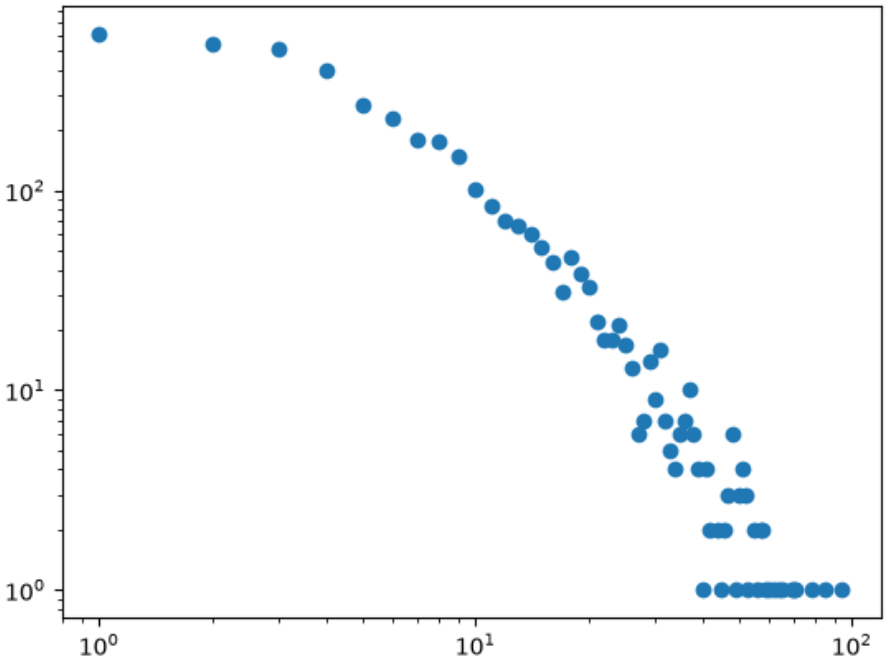


Figure 3.3: loss vs frequency

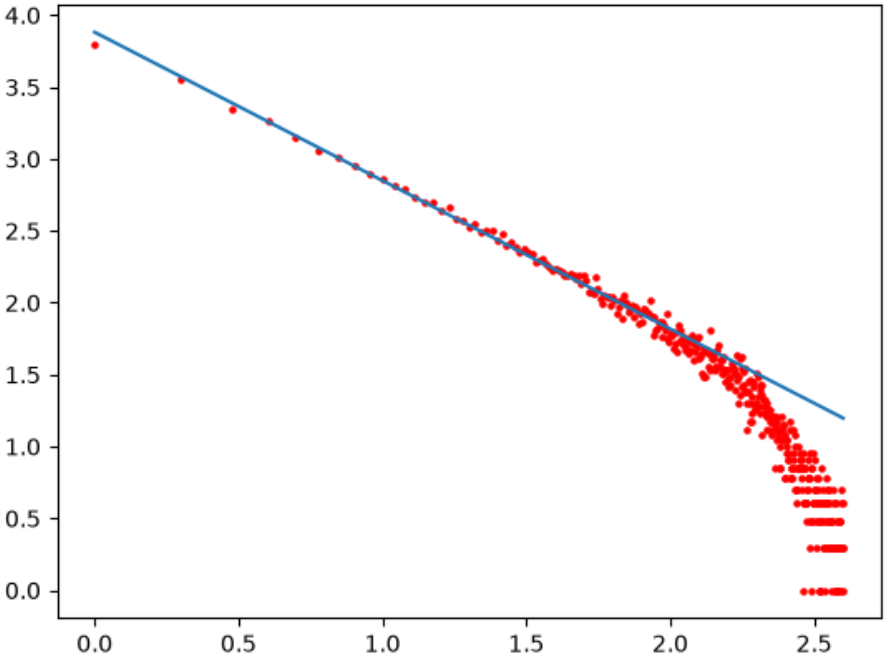


Figure 3.4: log topples vs log frequency

Chapter 4

Analysis

4.1 Analysis of the base model

4.2 Comparison of the extension to the base model

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